

# Botany Lessons





## About the Course

An elective in Life Science, Botany guides learners through the basic foundations of the discipline, while incorporating historical context, modern developments, current events, and citizenship. This course provides guidance and flexibility for teachers and learners to adjust the course for personal needs and preferences.

This topic is included in the following course(s): Science: Botany



## Placement & Combining Tips

There are no specific prerequisites or concurrents for this course.



## Scheduling

| GRADE | SCHEDULE INFO.                        | BOOKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-12  | Botany Lessons<br>5 times/week 45 min | A Peterson Field Guide To Western Trees<br>Botany in a Day<br>Growing a Revolution: Bringing Our Soil Back to Life<br>Pacific Coast Tree Finder<br>Rocky Mountain Tree Finder<br>Shanley's Quest Card Game<br>The Milkweed Lands: An Epic Story of One Plant: Its Nature and Ecology<br>Oaxaca Journal<br>Tree Finder - Eastern US<br>What a Plant Knows<br>Winter Tree Finder (Eastern US)<br>The Hidden Life of Trees: The Illustrated Edition<br>The Sibley Guide to Trees<br>Elementary Studies in Plant Life |

### Sample Weekly View

| Day 1                                                     | Day 2          | Day 3          | Day 4                                             | Day 5                         |
|-----------------------------------------------------------|----------------|----------------|---------------------------------------------------|-------------------------------|
| <b>Science: Botany</b>                                    |                |                |                                                   |                               |
| Botany Lessons<br>Nature Walks &<br>Scouting: Grades 9-12 | Botany Lessons | Botany Lessons | Botany Lessons<br>Nature Notebook:<br>Grades 9-12 | Botany Lessons<br>Botany Labs |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Botany Lessons

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

☐ Bookmark or print Quick Links, as needed.

## Term Prep & Teacher Tips

### Botany Lessons

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- various plants specimens, found or obtained as detailed in Elementary Plant Studies
- lily flower (Term 1)
- found twigs and fruits/seeds
- needle and/or blade for dissecting
- forceps/tweezers for dissecting
- large, transparent cup/glass
- masking tape
- aluminum foil
- 2 glass jars
- salt
- oil
- microscope coverslips and slides
- paper towels
- saucer
- recycled box
- pins
- cotton ball
- various corks
- recycled measuring spoon/cup marked in tsp/Tbsp or mL
- recycled apple corer or trowel
- place to dig soil
- distilled water
- zip top sandwich bags for soil samples
- small jars with secure lid for each soil sample
- dropper
- pea/bean seeds
- potting materials
- denatured alcohol
- iodine
- potato
- matches
- candle
- red food coloring/dye
- straw
- materials by student choice
- optional: tulip bulb

## Reminders

### Botany Lessons

☐ Note that Lab in week 7 requires a lily flower. Make a calendar note to obtain this closer to the appropriate time.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Botany Lessons



A Peterson Field Guide To Western Trees



Botany in a Day



Growing a Revolution: Bringing Our Soil Back to Life



Pacific Coast Tree Finder



Rocky Mountain Tree Finder



Shanleya's Quest Card Game



The Milkweed Lands: An Epic Story of One Plant: Its Nature and Ecology



Oaxaca Journal



Tree Finder - Eastern US



What a Plant Knows



Winter Tree Finder (Eastern US)



The Hidden Life of Trees: The Illustrated Edition



The Sibley Guide to Trees



Elementary Studies in Plant Life



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

### Botany Lessons

∞ [Seek app from iNaturalist](#)

∞ [Nature Journal Connection](#)

∞ [Alveary Bookshelf](#)

∞ [Elementary Studies in Plant Life](#)

∞ [Elementary Studies in Plant Life Annotations](#)

∞ [AmScope key for slide IDs](#)

Click THIS text  
or scan the QR  
code for links.



# Botany Lessons

## How To Approach



### Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



### Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



### Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



### Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



### Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

# Botany Lessons

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 1 45m Botany Lessons - Lesson 1

#### *Botany Literature*

Materials: What a Plant Knows

#### → INTRO

At least one day each week will be spent reading important works of botanical literature. The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.3-17.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.
- ☐ science free read

### WEEK 1 45m Botany Lessons - Lesson 2

#### *How Plants Work, Life Cycle of Flowering Plants*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.1. Work through the material, adding drawings to your notebook. You may need to visit local nurseries, botanical gardens, neighbors' yards, and gardens to find specimens to examine. Pick a question at the end to answer fully in your notebook for your narration.

### WEEK 1 45m Botany Lessons - Lesson 3

#### *Identification*

Materials: Botany in a Day

#### → READ, NARRATE, & DISCUSS

Forward, Region Covered, and p.1-9

#### → NOTE

You may or may not agree with the author's presentation of the evolution of plants. Still, it is important to know current botanical thinking, especially as we seek to understand how plants have been organized. This will help us as we begin to learn plant families.

### WEEK 1 45m Botany Lessons - Lesson 4

#### *Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → INTRO

Use this flex day to finish the week's reading from What a Plant Knows, or continue any lab/field activities. Suppose you have completed the week's activities. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 1

### WEEK 1 ☐ 45m Botany Lessons - Lesson 5

#### *Special Study*

☐ Materials: The Milkweed Lands

#### → INTRO

Notice how the author and illustrator make a study of and learn so much about this one plant. Whether you have milkweed near you and can notice their observations, or you make a similar study of a particular plant in your locale, enjoy their method as you develop your own.

#### → READ, NARRATE, & DISCUSS

Half of Ch.1 (to the end of "Dangerous Colors" p.8)

### WEEK 2 ☐ 45m Botany Lessons - Lesson 6

#### *Botany Literature*

☐ Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.17-31.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.

☐ science free read

### WEEK 2 ☐ 45m Botany Lessons - Lesson 7

#### *How Plants Work, Use of Water*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.2

### WEEK 2 ☐ 45m Botany Lessons - Lesson 8

#### *Identification*

☐ Materials: Botany in a Day

#### → READ, NARRATE, & DISCUSS

p.10-18

### WEEK 2 ☐ 45m Botany Lessons - Lesson 9

#### *Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows, or continue any lab/field activities. Suppose you have completed these activities for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 2 ☐ 45m Botany Lessons - Lesson 10

#### *Special Study*

☐ Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Finish Ch.1 from p.8.



## Term 1

### WEEK 3 45m Botany Lessons - Lesson 11

#### *Botany Literature*

Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.31-45.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.
- ☐ science free read

### WEEK 3 45m Botany Lessons - Lesson 12

#### *How Plants Work, Soil Science*

Materials: Links

#### → INTRO

Scientists now know that soil science is fundamentally important to plant growth and even human health. This is because of the vast number of organisms that live in the soil food web. Look at a basic diagram of this web at the link and copy it into your Botany Notebook. Then, watch the videos to consider how it works and why it's important.

#### → VIEW

∞ Image Link: Soil Food Web

#### → VIEW, NARRATE, & DISCUSS

∞ Video Link: The Living Soil Beneath Our Feet (2:49)

∞ Video Link: How Regenerative Agriculture Brings Life Back (15:22)

### WEEK 3 45m Botany Lessons - Lesson 13

#### *Identification*

Materials: Botany in a Day, Shanleya's Quest card game

#### → READ, NARRATE, & DISCUSS

p.19-21

#### → VIEW

Watch the author explain further.

∞ Link: Tutorial

∞ Link: Written Explanation in larger print from the website

#### → PLAY

Find a friend and begin playing the card games to familiarize yourself with the families and recognize the patterns.

### WEEK 3 45m Botany Lessons - Lesson 14

#### *Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose you have completed these activities for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



# Botany Lessons

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## Term 1

### WEEK 3 45m Botany Lessons - Lesson 15

*Special Study*

Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Half of Ch.2 (to p.27)

### WEEK 4 45m Botany Lessons - Lesson 16

*Botany Literature*

Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.46-59.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

Nature walk - Look for the plants and/or features mentioned in your books. Record observations.

science free read

### WEEK 4 45m Botany Lessons - Lesson 17

*How Plants Work, More Soil Science*

Materials: Links

#### → READ, NARRATE, & DISCUSS

Article Link: Soil Health

#### → VIEW

Video Link: Hopi Agriculture (4:12)

Video Link: Indigenous Farmer Solving Food Insecurity (16:11)

Video Link: Tiny Hooves Rewilding the Great Plains (13:25)

### WEEK 4 45m Botany Lessons - Lesson 18

*Identification*

Materials: Botany in a Day, Shanleya's Quest card game

#### → READ, NARRATE, & DISCUSS

p.79, 91, 109, 157, 163, 179, 192, and 207

#### → PLAY

Continue playing the card games until you are confident in your ability to recognize the 8 families.

### WEEK 4 45m Botany Lessons - Lesson 19

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 1

### WEEK 4 ☐ 45m Botany Lessons - Lesson 20

#### *Special Study*

☐ Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Finish Ch.2 from p.28.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 21

#### *Botany Literature*

☐ Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.60-73.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.

☐ science free read

### WEEK 5 ☐ 45m Botany Lessons - Lesson 22

#### *How Plants Work, Photosynthesis Basics*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.3

### WEEK 5 ☐ 45m Botany Lessons - Lesson 23

#### *Identification*

☐ Materials: Botany in a Day, Shanleya's Quest card game

#### → PRACTICE

Practice recognizing the 8 main families by playing the card games and observing plants. You will be introduced to a couple of new families each week. It might be possible to find all of them in the fall in your area. Ask local botanists at gardens, nature museums, or nature trails to assist you in your search. Read about and search for members of the Nymphaeaceae (p.50) and Lauraceae (p.53) families.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 24

#### *Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 25

#### *Special Study*

☐ Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Half of Ch.3 (to the end of p.47)

# Botany Lessons

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## Term 1

### WEEK 6 45m Botany Lessons - Lesson 26

*Botany Literature*

Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.73-86.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Begin collecting twigs and seeds for use in Term 2.
- ☐ science free read

### WEEK 6 45m Botany Lessons - Lesson 27

*How Plants Work, More on Photosynthesis*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.17

#### → VIEW

∞ Video Link: Photosynthesis Video (13:37)

### WEEK 6 45m Botany Lessons - Lesson 28

*Identification*

Materials: Botany in a Day

#### → READ

Read about and search for members of the Stonecrop (p.64) and Vitaceae (p.67) families.

### WEEK 6 45m Botany Lessons - Lesson 29

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 6 45m Botany Lessons - Lesson 30

*Special Study*

Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Finish Ch.3 (from p.48).

### WEEK 7 45m Botany Lessons - Lesson 31

*Botany Literature*

Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.86-102.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
- ☐ science free read

# Botany Lessons

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## Term 1

### WEEK 7 ☐ 45m Botany Lessons - Lesson 32

*How Plants Work, Flower Anatomy*

☐ Materials: Elementary Studies in Plant Life

→ **READ, NARRATE, & DISCUSS**

Ch.4

### WEEK 7 ☐ 45m Botany Lessons - Lesson 33

*Identification*

☐ Materials: Botany in a Day

→ **READ**

Read about and search for members of the Lythraceae (p.69) and the Onagraceae (p.70) families.

### WEEK 7 ☐ 45m Botany Lessons - Lesson 34

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

→ **NOTE**

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 7 ☐ 45m Botany Lessons - Lesson 35

*Special Study*

☐ Materials: The Milkweed Lands

→ **READ, NARRATE, & DISCUSS**

Half of Ch.4 (to the end of p.75)

### WEEK 8 ☐ 45m Botany Lessons - Lesson 36

*Botany Literature*

☐ Materials: What a Plant Knows

→ **READ, NARRATE, & DISCUSS**

Read about 14 pages per week; p.102-116.

→ **WRITE**

Write a narration in your Botany Notebook.

### WEEK 8 ☐ 45m Botany Lessons - Lesson 37

*How Plants Work, Plant Forms*

☐ Materials: Elementary Studies in Plant Life

→ **READ, NARRATE, & DISCUSS**

Ch.5

#### • **PLAN WEEKLY**

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
- ☐ science free read

# Botany Lessons

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## Term 1

### WEEK 8 45m Botany Lessons - Lesson 38

#### Identification

Materials: Botany in a Day

#### → READ

Read about and search for members of the Ulmaceae (p.87) and Cucurbitaceae (p.98) families.

### WEEK 8 45m Botany Lessons - Lesson 39

#### Flex Day

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 8 45m Botany Lessons - Lesson 40

#### Special Study

Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Finish Ch.4 (from p.76).

### WEEK 9 45m Botany Lessons - Lesson 41

#### Botany Literature

Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.116-129.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
- ☐ science free read

### WEEK 9 45m Botany Lessons - Lesson 42

#### How Plants Work, Leaf Anatomy

Materials: Elementary Studies in Plant Life, microscope, and prepared slides

#### → READ, NARRATE, & DISCUSS

Ch.6

- Using the prepared slide set, examine Slide 12, the Hydrilla verticillata whole mount. Notice the form of the leaf, the features along the margin, the tiny cells that repeat over and over, and anything else that stands out. You will learn more about cells in General Biology. Also, examine Slide 1, the pine leaf cross-section. A cross-section is a slice perpendicular to the length. Find the outer layer, called the epidermis, and the tiny pores through the epidermis that allow the leaf to breathe, called the stomata. Record what you see in your notebook.

# Botany Lessons

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## Term 1

### WEEK 9 ☐ 45m Botany Lessons - Lesson 43

#### *Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and search for members of the Cactaceae (p.123) and Ericaceae (p.132) families.

### WEEK 9 ☐ 45m Botany Lessons - Lesson 44

#### *Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 9 ☐ 45m Botany Lessons - Lesson 45

#### *Special Study*

☐ Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Half of Ch.5 (to p.99)

### WEEK 10 ☐ 45m Botany Lessons - Lesson 46

#### *Botany Literature*

☐ Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.129-143.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.

☐ science free read

### WEEK 10 ☐ 45m Botany Lessons - Lesson 47

#### *How Plants Work, Root Anatomy and Physiology*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.7

### WEEK 10 ☐ 45m Botany Lessons - Lesson 48

#### *Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and search for members of the Apocynaceae (p.140) and Convolvulaceae (p.145) families.

# Botany Lessons

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## Term 1

### WEEK 10 ☐ 45m Botany Lessons - Lesson 49

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 10 ☐ 45m Botany Lessons - Lesson 50

*Special Study*

☐ Materials: The Milkweed Lands

#### → READ, NARRATE, & DISCUSS

Finish the book (from p.100).

### WEEK 11 ☐ 45m Botany Lessons - Lesson 51

*Botany Literature*

☐ Materials: What a Plant Knows

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.144-163.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
- ☐ science free read

### WEEK 11 ☐ 45m Botany Lessons - Lesson 52

*How Plants Work, Trees and Shrubs in Winter*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.8

### WEEK 11 ☐ 45m Botany Lessons - Lesson 53

*Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and search for members of the Solanaceae (p.146) and Scrophulariaceae (p.150) families.

#### ★ TEACHER NOTE

You may want to have a few different types of flowers for the first week of Term 3! Pick some up from the florist that week, if needed.

### WEEK 11 ☐ 45m Botany Lessons - Lesson 54

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 1

### WEEK 11 45m Botany Lessons - Lesson 55

*Special Study*

Materials: Article Link

#### → READ, NARRATE, & DISCUSS

The naturalist's work is never done! We continually learn how much we don't know about plants, animals, and their relationships. Enjoy the article in the link!

∞ Article Link: South Carolina Swamps Harbor a Startling Secret

### WEEK 12 45m Botany Lessons - Lesson 56

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Botany Lessons - Lesson 57

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Botany Lessons - Lesson 58

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

#### ★ TEACHER NOTE

You may want to have a few different types of flowers for the first week of Term 3! Pick some up from the florist that week, if needed.

### WEEK 12 45m Botany Lessons - Lesson 59

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Botany Lessons - Lesson 60

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.



# Botany Lessons

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## Term 2

### WEEK 1 45m Botany Lessons - Lesson 61

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → INTRO

To ensure that the beautiful photography in this book is inspiring rather than distracting, take a few moments to look at the pictures on the pages you will read before beginning to read. The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, then you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.1-12.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- science free read

### WEEK 1 45m Botany Lessons - Lesson 62

*How Plants Work, Herbs in Winter*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.9

### WEEK 1 45m Botany Lessons - Lesson 63

*Identification*

Materials: Botany in a Day, a few different types of flowers

#### → READ, NARRATE, & DISCUSS

p.22-25. Begin to familiarize yourself with the keys on p.26-35 (you will have another week to do this). You may find it helpful to have a few different types of flowers to look at while examining the keys. The goal is to understand how they work and why we use them, rather than to memorize information.

#### → VIEW

The last 10 minutes of this video are on identifying new plants.  
∞ Video Link: Plant ID (38:50-46:03)

#### ★ STUDENT TIP

Finding flowers this time of year may require purchasing or asking for old ones at local florists. Once you get used to the way keys work, you will be ready to start using keys to ID trees that are around in winter. If you have trouble identifying a tree, move on to another tree, get help from local experts, use your Tree in Winter book, search online for local winter tree ID help, or find a local naturalist to ask.

### WEEK 1 45m Botany Lessons - Lesson 64

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 2

### WEEK 1 ☐ 45m Botany Lessons - Lesson 65

#### *Current Events*

☐ Materials: Links

#### → READ

Explore current events in Botany, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

∞ Link: The Smithsonian

∞ Link: The Botanical Society of America

∞ Link: Botany at Science Daily

∞ Link: The Plant Press

OR

#### → WRITE

Explore the intersection of science and faith, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

∞ Link: Veritas Forum

### WEEK 2 ☐ 45m Botany Lessons - Lesson 66

#### *Botany Literature*

☐ Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.13-27.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds

☐ science free read

### WEEK 2 ☐ 45m Botany Lessons - Lesson 67

#### *How Plants Work, Herbs in Winter*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.10

### WEEK 2 ☐ 45m Botany Lessons - Lesson 68

#### *Identification*

☐ Materials: Botany in a Day

#### → READ, NARRATE, & DISCUSS

p.22-25. Continue to familiarize yourself with the keys on p.26-35.

### WEEK 2 ☐ 45m Botany Lessons - Lesson 69

#### *Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are

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## Term 2

finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 2 45m Botany Lessons - Lesson 70

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 3 45m Botany Lessons - Lesson 71

*Botany Literature*

Materials: *The Hidden Life of Trees: The Illustrated Edition*

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.27-40.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds

science free read

### WEEK 3 45m Botany Lessons - Lesson 72

*How Plants Work, Twigs*

Materials: *Elementary Studies in Plant Life*, microscope, and prepared slides

#### → READ, NARRATE, & DISCUSS

Ch.15

• Using a set of prepared slides and the linked procedure, look at some plant stems under the microscope. Record your work in your notebook.  
∞ Link: Botanical Histology

#### → VIEW

Watch this video on the internal anatomy of stems and roots.

∞ Video Link: Stem and Root Anatomy (11:53)

### WEEK 3 45m Botany Lessons - Lesson 73

*Identification*

Materials: Link

#### → VIEW

Watch this video on tree ID in winter for a few tips. Practice ID'ing trees.

∞ Video Link: Tree ID (15:33)

### WEEK 3 45m Botany Lessons - Lesson 74

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are

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## Term 2

finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 3 45m Botany Lessons - Lesson 75

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 4 45m Botany Lessons - Lesson 76

*Botany Literature*

Materials: *The Hidden Life of Trees: The Illustrated Edition*

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.40-56.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- science free read

### WEEK 4 45m Botany Lessons - Lesson 77

*How Plants Work, Twigs*

Materials: Links

#### → PRACTICE

The two resources listed below are a very good introduction to how to use a key for identification. Study one or both to become comfortable with how they work. Then work to produce a twig key unique to your area. The twigs you have found and observed will be helpful.

#### → VIEW

- ∞ Link: Winter Tree ID Resource (a printable key)
- ∞ Link: Winter Twig Key Example (a web-based key)

### WEEK 4 45m Botany Lessons - Lesson 78

*Identification*

Materials: *Botany in a Day*, *The Sibley Guide to Trees*

#### → READ

Read about the Fagaceae family on p.100 of *Botany in a Day*. Members of the 3 Beech family genera can be compared in *The Sibley Guide* on p.172-226. Which of these is most widespread in North America? What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about.

### WEEK 4 45m Botany Lessons - Lesson 79

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from *The Hidden Life of*

# Botany Lessons

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## Term 2

Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 4 45m Botany Lessons - Lesson 80

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 5 45m Botany Lessons - Lesson 81

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.56-68.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds

science free read

### WEEK 5 45m Botany Lessons - Lesson 82

*How Plants Work, Twigs*

#### → PRACTICE

Work on your twig key.

### WEEK 5 45m Botany Lessons - Lesson 83

*Identification*

Materials: Botany in a Day, The Sibley Guide to Trees

#### → READ

Read about the Betulaceae family on p.101 of Botany in a Day. Members of the 5 Birch family genera can be compared in The Sibley Guide on p.150-171. What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

### WEEK 5 45m Botany Lessons - Lesson 84

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

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## Term 2

### WEEK 5 45m Botany Lessons - Lesson 85

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 6 45m Botany Lessons - Lesson 86

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.68-81.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds

science free read

### WEEK 6 45m Botany Lessons - Lesson 87

*How Plants Work, Twigs*

#### → PRACTICE

Work on your twig key.

### WEEK 6 45m Botany Lessons - Lesson 88

*Identification*

Materials: Botany in a Day, The Sibley Guide to Trees

#### → READ

Read about the Sapindaceae family on p.102-103 of Botany in a Day. Members of 2 Soapberry family genera can be compared in The Sibley Guide p.324-351. What family characteristics do these genera share? What distinguishes them? Find 2-3 species in the family located in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

### WEEK 6 45m Botany Lessons - Lesson 89

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 6 45m Botany Lessons - Lesson 90

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

# Botany Lessons

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## Term 2

### WEEK 7 45m Botany Lessons - Lesson 91

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.83-98.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read

### WEEK 7 45m Botany Lessons - Lesson 92

*How Plants Work, Fruits and Seeds*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.24

### WEEK 7 45m Botany Lessons - Lesson 93

*Identification*

Materials: Botany in a Day, The Sibley Guide to Trees

#### → READ

Read about the Pinaceae family on p.46 of Botany in a Day. Compare members of the Pine family in The Sibley Guide p.4-59. What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

### WEEK 7 45m Botany Lessons - Lesson 94

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 7 45m Botany Lessons - Lesson 95

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.



## Term 2

### WEEK 8 45m Botany Lessons - Lesson 96

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.98-108.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 8 45m Botany Lessons - Lesson 97

*How Plants Work, How Fruits and Seeds Spread*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.25

### WEEK 8 45m Botany Lessons - Lesson 98

*Identification*

Materials: Botany in a Day, The Sibley Guide to Trees

#### → READ

Read about the Magnolia (p.52) and Juglandaceae (p.99) families in Botany in a Day. Members of the Magnolia family can be compared in The Sibley Guide p.92-100. Members of the Walnut family can be compared on p.136-149.

OR

Read about the Cupressaceae family on p.48 of Botany in a Day. Members of the Cypress family can be compared in The Sibley Guide p.60-77.

- Find 2-3 species in the family located in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

### WEEK 8 45m Botany Lessons - Lesson 99

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 8 45m Botany Lessons - Lesson 100

*Current Events*

Materials: Links

#### → READ/WRITE



# Botany Lessons

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## Term 2

Continue exploring the links from Week 1.

### WEEK 9 45m Botany Lessons - Lesson 101

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.108-123.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 9 45m Botany Lessons - Lesson 102

*How Plants Work, Fruits and Seeds*

Materials: The Sibley Guide to Trees

#### → READ, NARRATE, & DISCUSS

"Fruit" p.xxxiii

#### → PRACTICE

Begin next week's practice, if time remains.

### WEEK 9 45m Botany Lessons - Lesson 103

*Identification*

Materials: The Sibley Guide to Trees

#### → READ

Consider a tree or shrub in your area that has not been studied yet this term. Use the Table of Contents in the front or the Index in the back of the Sibley Guide to find and read about its family. What characteristics do the different genera share? What distinguishes them? Read about at least 2 other family members. Are there even more species in your local tree guide or on your local cooperative extension website?

### WEEK 9 45m Botany Lessons - Lesson 104

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 9 45m Botany Lessons - Lesson 105

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

# Botany Lessons

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## Term 2

### WEEK 10 45m Botany Lessons - Lesson 106

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.125-140.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 10 45m Botany Lessons - Lesson 107

*How Plants Work, Fruits and Seeds*

Materials: The Sibley Guide to Trees

#### → PRACTICE

Observe your collection of seeds. Can you categorize them according to the different types described in your guide? Can you identify the plant from which each one came? What type of habitat does the parent plant live in? Or in what habitat did you find the seed? Based on the features of the seed, what is its mechanism of spreading? How do you suppose this benefits the plant? You will have more time for this observation next week.

### WEEK 10 45m Botany Lessons - Lesson 108

*Identification*

Materials: The Sibley Guide to Trees

#### → READ

Look through the Table of Contents in the Sibley Guide and find a tree family that sounds interesting. Read about that family. What characteristics are unique to this family? Are there any members of this family near you? You can search your local cooperative extension if you don't see any in the Guide.

### WEEK 10 45m Botany Lessons - Lesson 109

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 10 45m Botany Lessons - Lesson 110

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

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## Term 2

### WEEK 11 45m Botany Lessons - Lesson 111

*Botany Literature*

Materials: The Hidden Life of Trees: The Illustrated Edition

#### → NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

#### → READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.140-158.

#### → WRITE

Write a narration in your Botany Notebook.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 11 45m Botany Lessons - Lesson 112

*How Plants Work, Fruits and Seeds*

Materials: The Sibley Guide to Trees

#### → PRACTICE

Observe your collection of seeds. Can you categorize them according to the different types described in your guide? Can you identify the plant from which each one came? What type of habitat does the parent plant live in? Or in what habitat did you find the seed? Based on the features of the seed, what is its mechanism of spreading? How do you suppose this benefits the plant?

### WEEK 11 45m Botany Lessons - Lesson 113

*Identification*

#### → OBSERVE

Practice ID'ing trees. Record your observations.

### WEEK 11 45m Botany Lessons - Lesson 114

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

### WEEK 11 45m Botany Lessons - Lesson 115

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 12 45m Botany Lessons - Lesson 116

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.



## Term 2

|                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 117</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to the course.</p> |  |
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 118</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to the course.</p> |  |
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 119</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to the course.</p> |  |
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 120</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to the course.</p> |  |



## Term 3

### WEEK 1 ☐ 45m Botany Lessons - Lesson 121

*Botany Literature*

☐ Materials: Oaxaca Journal

#### → INTRO

The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, then you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week, to p.15.

#### → DRAW

Make a large sketch of the region where the book takes place. A nice map is on the first page. Then, as you read, fill in interesting things that he describes. At the end, you will have a map that represents some of the mental images that you have of the area after reading his descriptive writing.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 1 ☐ 45m Botany Lessons - Lesson 122

*How Plants Work, Germination Part 1*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.11

#### → REFLECT

If you have already taken Biology, recall what you learned about why oxygen is necessary for cellular respiration. Reflect on why you think the author did not explain all the details in the process.

### WEEK 1 ☐ 45m Botany Lessons - Lesson 123

*Identification*

☐ Materials: Botany in a Day

#### → INTRO

For a study of Botany, springtime is the time of year we have been waiting for! There are a lot more flower families to look for outside!

#### → READ

Read about and look for Ranunculaceae (p.55) and Salicaceae (p.76) families.

### WEEK 1 ☐ 45m Botany Lessons - Lesson 124

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 3

### WEEK 1 ☐ 45m Botany Lessons - Lesson 125

#### *Current Events*

☐ Materials: Links

#### → READ

Explore current events in botany, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

∞ Link: The Smithsonian

∞ Link: The Botanical Society of America

∞ Link: Botany at Science Daily

∞ Link: The Plant Press

OR

#### → WRITE

Explore the intersection of science and faith, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

∞ Link: Veritas Forum

### WEEK 2 ☐ 45m Botany Lessons - Lesson 126

#### *Botany Literature*

☐ Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.15-30.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

☐ Nature walk - record observations

☐ science free read

### WEEK 2 ☐ 45m Botany Lessons - Lesson 127

#### *How Plants Work, Germination Part 2*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.12

### WEEK 2 ☐ 45m Botany Lessons - Lesson 128

#### *Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and look for Sapindaceae (p.102) and Cornaceae (p.124) families.

### WEEK 2 ☐ 45m Botany Lessons - Lesson 129

#### *Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

# Botany Lessons

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## Term 3

### WEEK 2 ☐ 45m Botany Lessons - Lesson 130

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 3 ☐ 45m Botany Lessons - Lesson 131

*Botany Literature*

☐ Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.30-44.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

☐ Nature walk - record observations

☐ science free read

### WEEK 3 ☐ 45m Botany Lessons - Lesson 132

*How Plants Work, Herbs in Spring*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.13

#### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

### WEEK 3 ☐ 45m Botany Lessons - Lesson 133

*Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and look for the Iridaceae (p.201) family. Compare it to the Liliaceae family that you already know about and figure out the difference.

### WEEK 3 ☐ 45m Botany Lessons - Lesson 134

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 3 ☐ 45m Botany Lessons - Lesson 135

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

# Botany Lessons

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## Term 3

### WEEK 4 45m Botany Lessons - Lesson 136

*Botany Literature*

Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.44-56.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 4 45m Botany Lessons - Lesson 137

*How Plants Work, Trees and Shrubs in Spring*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.14

### WEEK 4 45m Botany Lessons - Lesson 138

*Identification*

Materials: Botany in a Day

#### → READ

Read about and look for Violaceae (p.75) and Caryophyllaceae (p.114) families.

### WEEK 4 45m Botany Lessons - Lesson 139

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 4 45m Botany Lessons - Lesson 140

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 5 45m Botany Lessons - Lesson 141

*Botany Literature*

Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.56-70.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read



# Botany Lessons

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## Term 3

### WEEK 5 ☐ 45m Botany Lessons - Lesson 142

*How Plants Work, Alternative Ways of Obtaining Food*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.18

#### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 143

*Identification*

☐ Materials: Botany in a Day

#### → READ

Read about and look for the Arecaceae (p.184) family.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 144

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 5 ☐ 45m Botany Lessons - Lesson 145

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 6 ☐ 45m Botany Lessons - Lesson 146

*Botany Literature*

☐ Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.70-85.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 6 ☐ 45m Botany Lessons - Lesson 147

*How Plants Work, Growth*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.19

# Botany Lessons

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## Term 3

### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

#### WEEK 6 45m Botany Lessons - Lesson 148

##### Identification

### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

#### WEEK 6 45m Botany Lessons - Lesson 149

##### Flex Day

Materials: Selected botany literature OR lab/field supplies

### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

#### WEEK 6 45m Botany Lessons - Lesson 150

##### Current Events

Materials: Links

### → READ/WRITE

Continue exploring the links from Week 1.

#### WEEK 7 45m Botany Lessons - Lesson 151

##### Botany Literature

Materials: Oaxaca Journal

### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.89-100.

### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

#### WEEK 7 45m Botany Lessons - Lesson 152

##### How Plants Work, Plants in Dry Places

Materials: Elementary Studies in Plant Life

### → READ, NARRATE, & DISCUSS

Ch.20

### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

#### WEEK 7 45m Botany Lessons - Lesson 153

##### Identification

### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

# Botany Lessons

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## Term 3

### WEEK 7 ☐ 45m Botany Lessons - Lesson 154

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 7 ☐ 45m Botany Lessons - Lesson 155

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 8 ☐ 45m Botany Lessons - Lesson 156

*Botany Literature*

☐ Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.103-114.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Botany Lessons - Lesson 157

*How Plants Work, Plants in Wet Places*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Read the first half of Ch.21.

#### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

### WEEK 8 ☐ 45m Botany Lessons - Lesson 158

*Identification*

#### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

### WEEK 8 ☐ 45m Botany Lessons - Lesson 159

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for

# Botany Lessons

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## Term 3

the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 8 ☐ 45m Botany Lessons - Lesson 160

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 9 ☐ 45m Botany Lessons - Lesson 161

*Botany Literature*

☐ Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.114-127.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 9 ☐ 45m Botany Lessons - Lesson 162

*How Plants Work, Plants in Wet Places*

☐ Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Read the second half of Ch.21.

#### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

### WEEK 9 ☐ 45m Botany Lessons - Lesson 163

*Identification*

#### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

### WEEK 9 ☐ 45m Botany Lessons - Lesson 164

*Flex Day*

☐ Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 9 ☐ 45m Botany Lessons - Lesson 165

*Current Events*

☐ Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

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## Term 3

### WEEK 10 45m Botany Lessons - Lesson 166

*Botany Literature*

Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.127-140.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 10 45m Botany Lessons - Lesson 167

*How Plants Work, Flowering Parts*

Materials: Elementary Studies in Plant Life

#### → READ, NARRATE, & DISCUSS

Ch.22

#### → PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

### WEEK 10 45m Botany Lessons - Lesson 168

*Identification*

#### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

### WEEK 10 45m Botany Lessons - Lesson 169

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 10 45m Botany Lessons - Lesson 170

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 11 45m Botany Lessons - Lesson 171

*Botany Literature*

Materials: Oaxaca Journal

#### → READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.141-159.

#### → MAP WORK

Add to your sketch map.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

# Botany Lessons

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## Term 3

### WEEK 11 45m Botany Lessons - Lesson 172

*How Plants Work*

Materials: Links

#### → VIEW

Watch these videos on stomata. Then take a few moments to observe the new leaves on the trees. How do they compare to those observed in Term 1?

∞ Video Link: Inside a Leaf (2:52)

∞ Video Link: Stomata #2 (3:50)

### WEEK 11 45m Botany Lessons - Lesson 173

*Identification*

#### → OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

### WEEK 11 45m Botany Lessons - Lesson 174

*Flex Day*

Materials: Selected botany literature OR lab/field supplies

#### → NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

### WEEK 11 45m Botany Lessons - Lesson 175

*Current Events*

Materials: Links

#### → READ/WRITE

Continue exploring the links from Week 1.

### WEEK 12 45m Botany Lessons - Lesson 176

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

#### • PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

### WEEK 12 45m Botany Lessons - Lesson 177

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Botany Lessons - Lesson 178

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

# Botany Lessons

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## Term 3

|                                                                                                                                                      |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 179</p> <p><i>Exam Week</i></p> <p>→ <b>EXAMS</b></p> <p>Answer question(s) related to the course.</p> |  |
| <p><b>WEEK 12</b> 45m Botany Lessons - Lesson 180</p> <p><i>Exam Week</i></p> <p>→ <b>EXAMS</b></p> <p>Answer question(s) related to the course.</p> |  |

# Botany Lessons

## Examination

### Term 1

#### • Day 1:

- What a Plant Knows: Pick one aspect of things that a "plant knows" and write about it.

#### • Day 2:

- Elementary Studies: Explain what you know about photosynthesis. Use sketches or diagrams.
- Elementary Studies: What are the different parts of a flower, and what are they used for? Use sketches or diagrams.

#### • Day 3:

- Botany in a Day: Outline with words and sketches the 8 common plant families we studied and how to recognize them.

#### • Day 4:

- The Milkweed Lands: Write a summary OR draw/diagram a 4-panel illustration of the ecological role that milkweed plays over the seasons.

#### • Day 5:

- Practical work or links: Describe with words and sketches your observations of 2 local plants OR discuss the importance of soil quality.

### Term 2

#### • Day 1:

- The Hidden Life of Trees: Pick one aspect of The Hidden Life of Trees and write about it.

#### • Day 2:

- Elementary Studies and/or practical work: Use words or pictures to contrast the form and function of at least two types of seeds OR describe the process of keying out a flower, twig, or tree.

#### • Day 3:

- Practical work: Use words and sketches to describe your observations of 2 local trees.

#### • Day 4:

- Current events: Discuss one current event in botany that you read about this term.

#### • Day 5:

- Practical work: Present your citizen science project to your teacher.

### Term 3

#### • Day 1:

- Oaxaca Journal: Describe some of the ferns the author saw in Oaxaca and some of his observations.

#### • Day 2:

- Elementary Studies: Describe the process of germination. How are a corn seed and a bean seed different?

#### • Day 3:

- Botany in a Day and practical work: What new plants have you seen locally since spring has arrived? How many of the 8 first families that we learned have you found?

#### • Day 4:

- Botany in a Day and practical work: Describe how you prepared soil samples for microscopic examination and explain any observations you made between different locations or times of year.

#### • Day 5:

- Practical work: Show your teacher your Botany Notebook. Select a favorite page to tell about and scan.